Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Software engineer with 8 years of production experience at Entrata, a PropTech company serving 100K+ monthly active users across affordable housing and property management. Owned business-critical data pipelines (HUD income limits, LIHTC eligibility, RabbitMQ async processing) from design through production incidents across 120+ Agile sprints. Now building at the intersection of software engineering and AI: **AgentProbe**, an open-source LLM evaluation framework with a CI-integrated 5-metric eval engine, and **Leasel**, an 8-agent LangGraph agentic platform for affordable housing compliance. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Looking for mid-level to senior individual contributor roles in software engineering, AI engineering, or applied data - particularly where domain expertise in real estate, housing, or regulated data adds genuine value.

WHAT I BRING

8 Years Production Software Engineering | Agentic AI · LangGraph · LLM Evaluation | Python · JavaScript · TypeScript · SQL | PropTech · Affordable Housing · HUD / LIHTC | RAG Pipelines · Data Pipelines · AWS | MS Data Analytics Engineering · GPA 3.87 | 120+ Agile Sprints · Mentoring · Code Reviews

WORK EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern (Unpaid)

- Built an agentic mission-planning system on LangGraph - parsed natural-language survey requests into structured flight plans, implemented a RAG layer over FAA Part 107 regulations for cited compliance answers, and enforced deterministic guardrails (altitude limits, battery budgets, no-fly-zone checks) before MAVLink export.

George Mason University - College of Engineering

Aug 2025 - May 2026 · Part-time

Graduate Teaching Assistant - MIS 303

- Delivered weekly workshops and one-on-one debugging sessions to 600+ students across multiple sections - Python programming, data structures, OOP, and database design (ER modeling, SQL, MS Access). Graded assignments with rubric-aligned feedback; coordinated content delivery with lead instructor.

George Mason University - Research

Aug 2024 - Aug 2025

Research Associate

- Built multi-source statistical models (Python, R) on WHO and UK Road Safety datasets across 190+ countries - $R^2 = 0.977$ regression accuracy; applied Isolation Forest for automated data integrity checks across 3 research pipelines; produced Tableau, ggplot2, and Quarto dashboards for faculty leadership (40% comprehension improvement across two cohorts).

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024 · Remote

Senior Software Engineer

- **Owned the HUD income-limit pipeline for 7 years** - transformed annual federal source data into ~180,000 domain records powering eligibility, rent, HAP, and LIHTC calculations for 100K+ housing units. Every release included baseline validation, reconciliation, and rollback preparation. Zero compliance violations across the entire ownership period.
- **Built and maintained RabbitMQ-based async processing** for gross-rent updates across thousands of tenant records - retry logic and idempotency guarantees that eliminated double-processing; owned the system through production incidents and annual rent-schedule changes.
- **Shipped 20+ full-stack production modules** (Python, PHP, PostgreSQL, JavaScript, TypeScript) across 120+ Agile sprints; led microservices redesign with Docker - 50% throughput improvement, 30% deployment time reduction, 80% reduction in manual operational workload through automation.
- **Optimized 200+ SQL queries (30% performance gain)** through execution plan analysis, indexing, partitioning, and stored procedure refactoring across multi-tenant schemas for eligibility, financial, and document workflows.
- **Led AWS S3 document migration** - 100% PHI compliance with Libsodium-encrypted pipelines; zero security incidents across the full migration.
- Mentored 5+ engineers; conducted 300+ peer code reviews; built debugging tools that reduced debugging time 40% and kept open bugs below 10 for 6+ consecutive months.

Entrata India (Db Xento Systems) - Pune

Jan 2017 - Sep 2018

Associate / Trainee Software Engineer

- Built affordable housing data modules (income limits, 1.6M HUD/Tax Credit records across 5 program years), document management, and compliance workflows in PHP and Python. Led cloud migration to AWS S3; redesigned document storage database architecture for 180K+ records.

PROJECTS

AgentProbe: Open-Source LLM Evaluation Framework

Open Source

Python · LangGraph · FastAPI · ChromaDB · CI/CD (GitHub Actions) · 2025

github.com/venkatalakshmiparimala/AgentProbe

A fully local, zero-cost AI agent evaluation toolkit: 15-endpoint FastAPI backend, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate that blocks quality regressions before merge, and multi-provider LLM routing (OpenAI, Anthropic, Groq, Ollama). Any agent connects via REST API without code changes. Validated LeaseI across 1,240+ compliance test cases - pass@1 81.7%, pass@3 88.3% via multi-trial sampling harness with LLM-as-judge scoring.

LeaseI: Agentic AI Platform for Affordable Housing Compliance

Python · LangGraph (8-agent) · FastAPI · Streamlit · TF-IDF RAG · OCR · SQLite · HUD API · 2026

github.com/venkatalakshmiparimala/Lease_complaince_rag

Three-surface enterprise knowledge platform (Document Intelligence, Support Investigation, Tenant Self-Service) sharing one SQLite repository. The support investigation agent is an 8-node LangGraph StateGraph - Triage, Retrieval, Configuration, Historical Incident, Structured Data, Reasoning, Validation, Human Escalation - with real conditional edges and HITL when confidence is too low. Hand-rolled TF-IDF RAG (no LangChain, no vector DB) over 1,240 residential leases and federal HUD/LIHTC regulations. Core design rule enforced in code: LLM reasons; human or deterministic logic makes every final eligibility and compliance call. Results: 93.6% precision, 87.9% F1, 100% citation correctness, 0% unsupported-claim rate.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · React · LLMs · FHIR API · Docker · spaCy · BERT · Embeddings · 2024

github.com/venkatalakshmiparimala/AI_on_FHIR

Flask + React AI platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings. Reduced manual clinical retrieval time ~60%; enforced HIPAA-compliant RBAC across structured medical records. Containerized with Docker.

CrashClarity: Road Safety Statistical Analysis (104K Records)

Python · R · scikit-learn · Random Forest · Logistic Regression · GitHub Pages · 2025

venkatalakshmiparimala.github.io/Portfolio

End-to-end analysis of 104K+ UK road accident records - Random Forest + Logistic Regression ensemble with feature engineering across weather, road, and temporal variables (85%+ accuracy); interactive data visualization dashboard deployed on GitHub Pages.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering - Jawaharlal Nehru Technological University, Anantapur — 2017

VOLUNTEER

Saving Hands NGO - Pune, India — 2018 - 2024

Volunteer Educator

- Provided academic support and exam preparation to visually impaired children over 6 years, concurrent with full-time employment. Adapted instructional approaches to individual learning needs; participated in health checkups and fundraising.

SKILLS

Languages: Python, JavaScript, TypeScript, PHP, SQL, PL/SQL, R, Bash, HTML/CSS

AI & Agentic: LangChain, LangGraph, RAG Pipelines, Multi-Agent Orchestration, LLM Evaluation, Prompt Engineering, HITL, ChromaDB, Vector Databases, Embeddings, OpenAI API, Anthropic API, FastAPI, MCP

ML & Data: PyTorch, scikit-learn, XGBoost, Random Forest, Logistic Regression, Isolation Forest, PySpark, NumPy, Pandas, Tableau, Power BI, ggplot2, Quarto, spaCy, BERT, Hugging Face

Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), RabbitMQ, Microservices, PostgreSQL, MySQL, Oracle, Git, Linux, Agile/Scrum